



Natural gas reforming in thermochemical waste-heat recuperation systems: A review

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ABSTRACT

This article summarizes researches on thermochemical waste-heat recuperation (TCR) systems based on natural gas reforming. It includes the results of an industrial application, experimental lab tests, and simulation of TCR for various fuel-consuming equipment. A classification of thermochemical recuperation systems is proposed. In this review, thermochemical recuperation of different types of waste-heat such as exhaust gas, molten slag, elements of thermal protection has been considered. The TCR systems based on various reactions of methane reforming (with steam, carbon dioxide, a mixture of steam and carbon dioxide, steam and flue gases, a mixture of steam and oxygen) are summarized. After analyzing the articles on thermochemical recuperation, a flowchart of tasks for the TCR study was proposed. This flowchart can virtually be divided into six blocks: thermodynamic analysis; TCR configuration; heat and mass transfer in the reformer; catalysts design; synthesis gas combustion; experimental testing and industry application. The analyzed TCR systems perform the traditional function of ensuring waste-heat recovery and allow the production of fuel with superior characteristics containing molecular hydrogen. As a result, it becomes possible to use primary hydrocarbon fuel and hydrogen-rich gas – products of endothermic reforming. Considering, the global trend of using hydrogen as an energy source, the production of hydrogen at the equipment where it is consumed (on-board hydrogen production system) could use the TCR systems with endothermic hydrocarbon fuel reforming. According to this statement, analysis has been made on the impact of TCR on the emission of NO_x and CO_2 , when synthesis gas after TCR systems is used. TCR based on steam methane reforming can be considered as ultralow NO_x emission technology for gas turbine application.

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1. Introduction

Sustainable development of the world economy is inextricably linked with energy. At the same time, in the last century, fossil fuels have played a leading role in the energy sector. In the 21st century, despite advances in alternative and renewable energy sources, the share of fossil fuels in the global energy balance (according to World Energy Balance Highlights 2020 Edition presented by the International Energy Agency (IEA)) is about 64% [1]. According to the same highlights, the share of fossil fuels has decreased by 3% over the past ten years. However, while the production of oil and coal (with peat and oil shale) has changed insignificantly over ten years, the natural gas production has increased from 2712 Mtoe to 3632 Mtoe. In addition, the IEA report “Golden Rules for a Golden Age of Gas” shows that natural gas (together with shale gas) will be the main source of energy in the 21st century for the global economy [2].

The widespread use of fossil fuels in modern energy can be explained by the following reasons:

- advanced production and use technologies;
- well-established logistics from a producer of fossil fuels to consumer;
- relatively low cost.

The above arguments and facts allow us to reasonably believe that fossil fuels and, in particular, natural gas will be the main source of energy in the coming decades. Therefore, efficient improvement of the use of fossil fuel will have wide attention of the engineers and researchers worldwide in the near future.

Some fuel-consuming equipment that uses fossil fuels has almost maximum energy efficiency (for example, steam and water boilers). However, for a large number of fuel-consuming equipment, the coefficient of fuel heat utilization is much less than unity. Such equipment includes industrial furnaces, high-temperature plants, gas turbines, internal combustion engines, etc. Therefore, the task of increasing the efficiency of such fuel-consuming equipment is critically important at the present time.

Among the ways to improve efficiency in the fuel-consuming equipment for various purposes (heat technology, heat and power, etc.) there exist two main approaches. The first is the rational use of conventional fuel by improving the fuel and the technological cycles [3,4]. The second approach is the replacement of the original fuel with an alternative one combined with other energy resources [5–7]. One way to increase energy efficiency, which employs the two ways mentioned above, is thermochemical waste-heat recuperation (TCR) [8–11]. The main concept of thermochemical recuperation is transformation of exhaust gases heat into chemical energy of a new synthetic fuel that has higher calorific properties such as low-heating value [12]. This new synthetic fuel produced in the reformer is used in the combustion chamber of fuel-consuming equipment, reducing specific consumption of the primary fuel. The heat losses are the main reason for the low energy efficiency of a wide range of fuel-consuming equipment. Therefore, the recovery of waste-heat to the technological cycle of such installations is a crucial task for engineers and researchers.

The TCR system performs not only the traditional function of ensuring heat recovery but also allows the production of fuel with

superior characteristics containing molecular hydrogen [13–15]. As a result, it becomes possible to use not only primary hydrocarbon fuel but also hydrogen-rich gas – a product of endothermic hydrocarbon fuel reforming [16]. Taking into account the global trend of using hydrogen as an energy source [17–19], the TCR system with endothermic hydrocarbon fuel reforming could be utilized for producing hydrogen at a plant where it is consumed (on-board hydrogen production system) [20].

Recently, various articles on thermochemical waste-heat recuperation have been published. Some researchers consider thermochemical recuperation to be one of the most promising ways to improve the energy and environmental efficiency of fuel-consuming equipment [8,11,21,22].

The thermochemical waste-heat recuperation systems can be classified according to the following diagram presented in Fig. 1 based on the following criteria: type of fuel, type of oxidizing agent, and temperature of waste-heat. Mainly three types of fuel can be used for TCR: gaseous (natural gas, propane), liquid (ethanol, methanol, diesel), solid (various types of coal). Steam is one of the most common oxidizing agents for TCR. However, carbon dioxide, as well as flue gases, can be used as the oxidizing agent for TCR. The efficiency of the thermochemical recuperation systems strongly depends on the operation parameters (in particular, temperature). Therefore, the TCR systems can be classified based on the temperature of waste-heat (exhaust gas): high-temperature thermochemical recuperation system (above 700 °C), moderate temperature (300–700 °C), low temperature (100–300 °C). One of the main criteria of choosing the endothermic process for TCR is temperature [23].

This review is intended to systematize and summarize the available articles on TCR based on reforming of natural gas (methane), as well as to show the prospects for the further development of the TCR technology. Recent advances in processing methods and some of the key applications are reviewed in greater detail below. In addition, based on the author's previous researches, and the articles of other authors, a scheme of tasks for the development of TCR was drawn up.

2. Historical retrospective

Before we consider and analyze the various thermochemical recuperation schemes via reforming of natural gas (methane), a short historical retrospective will be given. Despite the fact that the reforming of hydrocarbons has been known for more than 100 years, scientific publications on thermochemical recuperation begin to be found only in the early 60s of the 20th century. Recently, Tartakovsky and Sheintuch presented a comprehensive historical

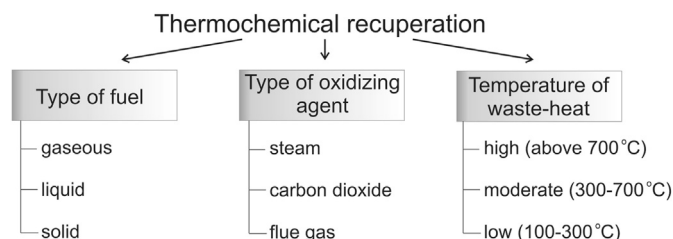


Fig. 1. A classification of the thermochemical waste-heat recuperation systems.

perspective on the use of fuel reforming in internal combustion engines (ICE) [8].

The idea of using hydrocarbon fuel reforming to increase the energy efficiency of fuel-consuming facilities was discussed as early as the 1960s [24]. Interestingly, Lee and Wimmer initially considered pre-combustion fuel reforming as a way to reduce pollutant emissions from spark-ignition engines. Lee and Wimmer [24] reported that the products of reforming of liquid hydrocarbon fuel with steam have better lean burning behavior and produced lower emissions than propane and methane. However, these authors used external heat sources for reforming of liquid hydrocarbon fuels. The main reason for better burning and emission characteristics is hydrogen and carbon monoxide in the reforming products.

The idea of using the reforming products as fuel in ICE was developed further. To obtain hydrogen-rich fuel, Houseman and Cernini [25] suggested using the gasoline partial oxidation process as an on-board hydrogen production technology. The authors considered gasoline partial oxidation over the Ni-based catalysts. These authors were the first who investigated the effect of the addition of the reforming products to the gasoline-air mixture in ICE. Houseman and Cernini considered the auto-thermal partial oxidation without external heat supply.

One of the first who suggested using exhaust heat of ICE for fuel reforming was Lindstrom [26]. However, the main goal of Lindstrom's invention was reducing the amount of polluting components in exhausts from ICE. And only a few years later, Martin [27] considered ICE with hydrocarbon fuel reformer as a total energy conversion system where the increase in ICE efficiency is due to the use of exhaust heat to reform the original fuel (gasoline). In this study, waste heat from the ICE was utilized for the steam gasoline reforming at about 900 K over the catalysts.

Further, the idea of using fuel reforming before combustion in an internal combustion engine found a wide discussion in the scientific literature. One of the latest works on the effect of reforming on the combustion and emission characteristics of ICE are the articles of specialists from the Technion Internal Combustion Engines Laboratory at the Technion-Israel Institute of Technology [28]. TCR via reforming of liquid hydrocarbon fuels was mainly discussed for ICE. However, there are a few works from the 60s and 70s of the last century on thermochemical waste-heat recuperation by natural gas reforming.

The earliest known work on thermochemical recuperation in an industrial gas furnace is the patent of Professor Schultz in 1961 [29]. Shultz suggested using the steam reforming of natural gas for waste-heat recuperation of continuous billet-heating furnace. He presented a schematic diagram where to carry out the steam natural gas reforming the heat of exhaust gases is used. Synthesis gas after reforming was used as a fuel for a continuous billet-heating furnace.

Further, the investigations on the industrial application of thermochemical waste-heat recuperation were continued in the USSR for improving the energy efficiency of glass-melting furnaces. A scientific group on TCR investigation led by Pereletov was created at the Moscow Power Engineering Institute in 1965. One of the main results of this scientific group was the creation and industrial application of the thermochemical waste-heat recuperation system for a glass-melting furnace [30]. These researchers showed that heat recuperation from waste flue gases can be improved by using the waste flue gases heat for natural gas reforming in a glass-melting furnace.

The earliest known work on TCR application in a gas turbine is a paper written by Olmsted and Grimes published in 1972 [31]. The authors suggested partially replacing the steam superheater section with a steam methane reformer. The exhaust heat after a gas turbine was used for the endothermic steam methane reforming

process. Then the products of reforming (synthesis fuel) (consists of hydrogen, carbon monoxide, carbon dioxide, excess steam, unreformed methane) are fed into the turbine combustor. In that paper, Olmsted and Grimes showed that $\text{CH}_4/\text{H}_2\text{O}$ absorbs exhaust heat thermally (as the temperature of mixture increases), and chemically (as the endothermic process takes place), resulting in a higher potential recuperation of exhaust heat than can be observed by conventional recuperation using air heating. Further, the idea of using TCR in gas turbines was found a wide discussion in the scientific literature. A detailed analysis of the gas turbines with TCR will be presented below.

Summing up the historical retrospective, we can say that the era of thermochemical waste-heat recuperation began in the 60s of the 20th century. In the 60s and 70s, the first experimental and industrial results of the use of thermochemical recuperation in internal combustion engines, industrial furnaces, and gas turbines were obtained. Over the next decades, interest in thermochemical recuperation remains high among scientists and engineers around the world.

3. TCR concept

The basic concept of TCR is based on the use of waste-heat for the endothermic reaction of hydrocarbon fuel reforming. A distinctive feature of fuel-consuming equipment with TCR is the fact that the transformation of the chemical energy of the fuel into heat occurs in two stages [10,32]. Whereas in traditional fuel-consuming equipment such transformation occurs in one stage (direct fuel combustion), in fuel-consuming equipment with TCR, the transformation of the chemical energy of the fuel into heat is divided into two stages. The first stage is endothermic reforming of the initial hydrocarbon fuel by using waste-heat. In this case, the combustion heat of the reforming products will be higher than the combustion heat of the initial fuel (per 1 kg of the initial fuel). The second stage is the combustion of reforming products as fuel.

Fig. 2 shows a comparison of the transformation of the chemical energy of fuel into heat in an industrial furnace without TCR and in an industrial furnace with TCR.

In the schematic diagram presented in Fig. 2 the steam reforming reaction is considered. However, as it will be shown below, the various oxidizing agents can be used in the thermochemical waste-heat recuperation systems. Also, it can be seen from this schematic diagram various hydrocarbon fuels can be used for TCR. The author's previous work [23] gives the answers to the question "How to choose an endothermic process for thermochemical waste-heat recuperation?" These answers are the following:

- initial fuel and oxidizing agent for TCR should have great availability and low cost;
- heat recuperation rate should tend to unity;
- heat transformation coefficient should have a maximum value.

Various hydrocarbon fuels can be used for the operation of the thermochemical waste-heat recuperation systems. A schematic diagram of the thermochemical waste-heat recuperation system has a similar structure for various fuel-consuming equipment: gas turbines, ICEs, industrial furnaces, etc. Fig. 3 shows a schematic diagram of fuel-consuming equipment with thermochemical waste-heat recuperation by reforming of hydrocarbon fuels. In this schematic diagram, two type of oxidizing agents is considered: steam and/or carbon dioxide. There are the solutions when flue gas (mainly containing a mixture of H_2O , CO_2 and N_2) are used as an oxidizing agent for fuel reforming [21,33,34].

Various chemical reactions take place in the reformer for a given

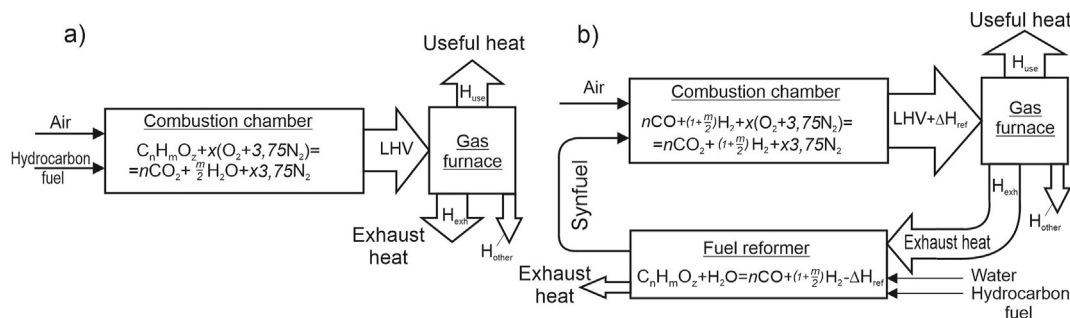


Fig. 2. Schematic diagram of the transformation of the chemical energy of hydrocarbon fuel into heat in an industrial furnace without TCR (a) and with TCR (b): LHV - low heating value, H_{other} - other heat losses, ΔH_{ref} - the enthalpy of hydrocarbon reforming process.

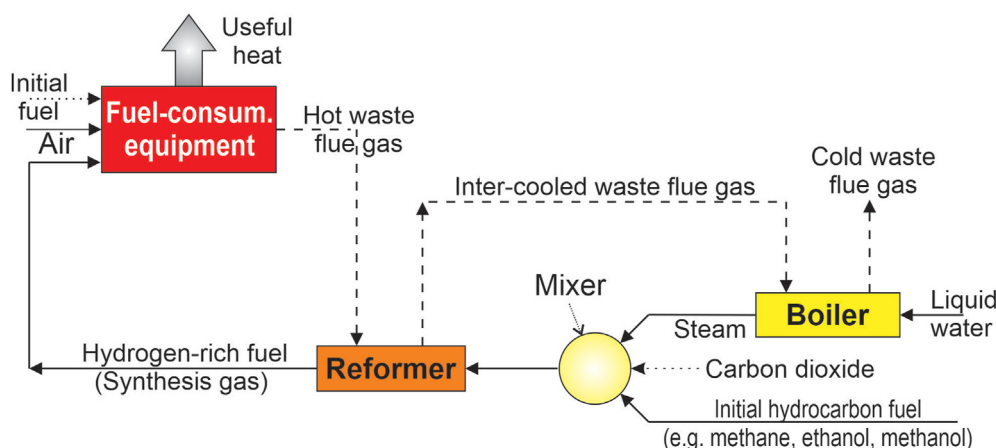


Fig. 3. A schematic diagram of fuel-consuming equipment with thermochemical waste-heat recuperation by reforming of hydrocarbon fuels.

endothermic process. For example, 11 main reactions take place in the reformer when steam reforming of methane is performed [35]. However, if we consider the resulting overall reactions for each hydrocarbon fuel and each oxidizing agent for stoichiometric conditions, the following tables (Table 1) can be obtained.

In Table 1 the following notations are adopted:

- LHV_{if} , MJ/kg – low heating value of 1 kg of initial hydrocarbon fuel;
- ΔH_{298}^0 , kJ/mol – enthalpy of reforming reaction;
- LHV_{syngas} , MJ/kg_{if} – low heating value of synthesis gas obtained by reforming of 1 kg of initial hydrocarbon fuel;
- η_q – heat transformation coefficient (HTC). HTC is the ratio of LHV_{syngas} and LHV_{if} .

Table 1

Comparison of various endothermic processes for thermochemical waste-heat recuperation.

Chemical reaction	LHV_{if} , MJ/kg	ΔH_{298}^0 , kJ/mol	LHV_{syngas} , MJ/kg _{if}	η_q
1. $CH_4 + H_2O \rightleftharpoons CO + 3H_2$ [36]	50.009	206.1	62.854	1.2568
2. $CH_4 + CO_2 \rightleftharpoons 2CO + 2H_2$ [37]	50.009	247.3	65.410	1.3070
3. $C_3H_8 + 3H_2O \rightleftharpoons 3CO + 7H_2$ [38]	46.357	499.1	57.674	1.1829
4. $C_3H_8 + 3CO_2 \rightleftharpoons 6CO + 4H_2$ [39]	46.357	620.3	60.418	1.3033
5. $CH_3OH \rightleftharpoons CO + 2H_2$ [40]	19.930	91.0	22.772	1.1426
6. $C_2H_5OH + H_2O \rightleftharpoons 2CO + 4H_2$ [41]	26.700	210.2	34.514	1.1565
7. $C_3H_5(OH)_3 \rightleftharpoons 3CO + 4H_2$ [42]	16.830	251.2	19.709	1.1710
8. $C_8H_{18} + 8H_2O \rightleftharpoons 8CO + 17H_2$ [43]	44.427	1310.0	55.895	1.2581

Table 1 shows that the maximum heat transformation coefficient is observed for two reactions of dry reforming of methane (2) and propane (3). The heat transformation coefficients for these reactions are above 1.3. The minimum heat transformation coefficient is observed in the steam ethanol reforming (6) and decomposition of methanol (5) and glycerol (7). Table 1 shows the stoichiometric reactions but these reactions are accompanied by a wide range of other side reactions. The synthesis gas composition after reforming or decomposition reactions depends on temperature, pressure, the composition of the inlet reaction mixture, type of catalysts (especially for reforming and decomposition of alcohols). The thermodynamic analysis can be used to determine the equilibrium product gas composition. However, the thermodynamic analysis is not allowed to calculate the effect of a catalyst and the type of reformer of the synthesis gas composition. For these goals, experimental investigation and CFD-modelling can be performed. The modern tools for CFD-modelling, such as ANSYS Fluent and CFX, Comsol Multiphysics, OpenFOAM, StarCCM++, Flow Fision, are powerful process simulators for modeling various steady-state and unsteady-state chemical processes.

In this review, the thermochemical waste-heat recuperation systems based on natural gas (methane) reforming will be considered in the detail. However, there are a lot of articles on other ways of TCR. The thermochemical waste-heat recuperation systems based on reforming of liquid fuels found a wide investigation for application in ICE. The latest papers on TCR by reforming of liquid hydrocarbon fuels have a wide variety. Poran et al. considered the TCR based on reforming of methanol [44–48]. Tartakovsky et al. performed comparative performance analysis of spark ignition

engine when synthesis gas after the methanol and ethanol reforming processes are used as a fuel [28,49–51]. Chakravarthy et al. carried out an investigation of the theoretical potential of thermochemical exhaust heat recuperation based on ethanol and isooctane for internal combustion engines [22]. Gaber et al. presented the results of thermochemical analysis and experimental investigation of a recuperative waste heat recovery system based on the tri-reforming of light oil [52]. Vernon et al. determined the potential for exhaust heat recovery by thermochemical recuperation based on ethanol steam reforming for the hydrogen-enriched internal combustion engines [53]. Cherednichenko and Serbin reported the results of energy analysis of the ship propulsion system with thermochemical waste-heat recuperation based on the diesel reforming process [54]. Qin et al. performed an experimental investigation on the thermochemical recuperation process based on *n*-decane reforming for scramjet cooling [55]. One of the promising ways for the use of thermochemical recuperation is the thermal protection of thermally stressed structures. Leung et al. proposed the use of thermochemical recuperation for raising the fuel heating value and recovering exhaust heat by on-board oxidative reforming of bioethanol [56]. Tola and Lonis proposed the gas turbine with thermochemical recuperation fed by renewable methanol [57]. Wang et al. presented the performance study on methanol steam reforming rib micro-reactor with waste heat recuperation [58]. Pan et al. developed the operational rules and control strategies for the application of thermochemical exhaust gas recuperation based on diesel steam reforming in the marine propulsion [59]. The study of using TCR in a port-injection spark-ignition four-cylinder natural gas engine was performed by Sun et al. [60].

The reactions of the coal gasification are also strongly endothermic. Therefore, there is an article on using the coal gasification processes for thermochemical waste-heat recuperation. Kuchonthara et al. presented the combination of thermochemical recuperative coal gasification cycle and fuel cell for power generation [61]. The authors proposed an integrated power generation cycle based on thermochemical recuperation by coal gasification and a solid oxide fuel cell. They showed that the cycle efficiency with TCR based on coal gasification increases from 39.5% (HHV) without the solid oxide fuel cell to about 45% (HHV) with the solid oxide fuel cell. Hayashi et al. proposed using the steam gasification of carbonaceous solid fuels for thermochemical recuperation to improve the efficiency of combined-cycle power generation and hydrogen production [62,63]. The authors analyzed various solid fuels, including biomass and lignite as well as various oxidizers such as steam-oxygen mixture and steam-air. Previously, the author of this review published a paper dialing with an application of thermochemical waste-heat recuperation by coal gasification in the industrial furnaces [64]. The main goal of this paper is a calculation of heat recuperation degree for various schematic diagrams of TCR by coal gasification. It was found that the maximum heat recuperation degree of 0.97 can be achieved at a steam-to-carbon ratio of 1 at a temperature of exhaust gases of 800 °C.

Recently, the necessity of reducing carbon emissions in fuel-consuming equipment has been reflected in many governmental policy documentation and roadmaps. One of the interesting ways of thermochemical waste-heat recuperation for carbon-free fuel-consuming equipment is TCR based on an ammonia decomposition process [65]. Sittichompoo et al. proposed the exhaust energy recovery system via catalytic ammonia decomposition for hydrogen production in low carbon clean vehicles [66]. The author showed that the products of ammonia decomposition contain up to 15% more chemical energy (low heating value) than initial ammonia. Wang et al. considered the ammonia decomposition process as on-board hydrogen production technology [67]. The authors proposed

the use of exhaust heat for an endothermic ammonia decomposition process.

Despite the greater variety of thermochemical waste-heat recuperation technologies, one of the most widespread technologies for TCR is TCR based on natural gas reforming.

4. TCR by natural gas reforming

A large number of studies have been carried out on thermochemical waste-heat recuperation via methane reforming. It is known that the degree of methane conversion tends to unity at temperatures above 800–900 K. Therefore, thermochemical waste-heat recuperation due to methane reforming is applicable primarily for high-temperature heat. In this regard, this section systematizes researches on TCR in industrial furnaces, gas turbines, etc.

Articles and patents on thermochemical recovery can be divided into three groups: industrial application studies, laboratory tests and simulation results. Therefore, this section contains three corresponding subsections.

4.1. Industrial application

Industrial furnaces generally have high flue gas temperatures. Table 2 shows the thermal performances of several industrial furnaces. In this table, waste-heat losses (%) are the relation of enthalpy of flue gases after the furnace body to the total enthalpy in the furnace body.

The main fuel for industrial gas furnaces is natural gas (the main component is methane). Moreover, the methane conversion of steam and carbon dioxide reforming at temperatures above 800 °C tends to unity. Therefore, the methane reforming processes (with steam, carbon dioxide, or flue gas) are widely used in the thermochemical waste-heat recuperation systems for waste-heat recovery in industrial gas furnaces.

The investigations of TCR by methane reforming can be virtually divided into three groups: industrial utilization, laboratory test, and theoretical (numerical) analysis. Among experimental investigations, it is important to highlight the investigations in the USSR where various research groups realized in the industry different thermochemical recuperation systems. As it was shown above, one of the first results of industrial utilization of the TCR system by steam methane reforming was a glass-melting furnace constructed by specialists from Moscow Power Engineering Institute in cooperation with Gusevsky fiberglass plant [69,70]. The schematic diagram of this glass-melting furnace with TCR is presented in Fig. 4.

In this schematic diagram (Fig. 4) the TCR system is used with a combination of the recuperation system by air heating. The hot waste flue gases are divided into two steam. The first stream is fed for natural gas reforming in a reformer. The inter-cooled flue gases after the reformer are used to preheat natural gas. In this schematic diagram, steam is produced outside the waste-heat recuperation system. The second stream of hot flue gases after a glass-melting furnace is passed through two stages of an air heater. Both

Table 2
The thermal performances of several industrial furnaces [68].

Type of furnace	Temperature in furnace body, °C	Flue gas temperature, °C	Waste-heat losses, %
Steel-making	1650–1750	1450–1550	65–75
Pit furnace	1350–1450	1200–1300	55–60
Forging	1200–1400	1000–1100	40–50
Glass-melting	1600–1700	1000–1100	50–60
Slab heating	1300–1400	900–1100	30–45

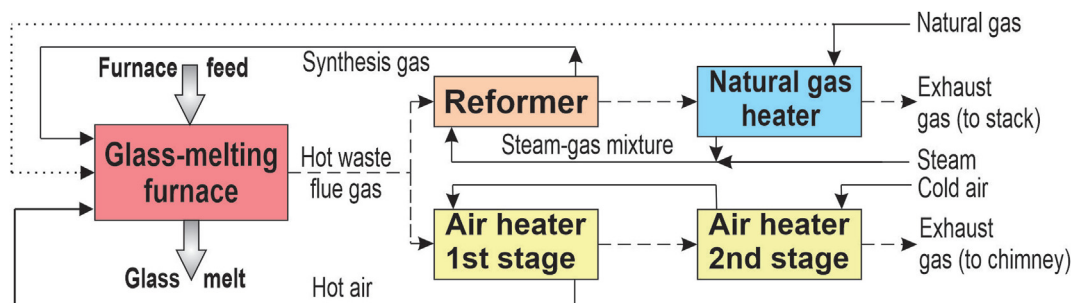


Fig. 4. The schematic diagram of that glass-melting furnace with TCR [69,70].

synthesis gas after the reformer and hot air after air heater is used for combustion in a glass-melting furnace.

Pereletov and co-worker showed that the specific fuel consumption per 1 kg of charge decreased from 0.74 to 0.524 m³, i.e. by 29.2%, or 1.41 times, when the furnace was operated with thermochemical recuperation compared to the furnace without thermochemical recuperation. They also showed experimentally that the specific fuel consumption per 1 kg of fiberglass decreased from 1.34 to 0.994 m³, i.e. by 25.8%, or 1.35 times. Thus, in a glass-making furnace with thermochemical recuperation, the maximum heat recuperation degree was 0.621. Table 3 shows the results of industrial testing of a glass-melting furnace without TCR and with TCR.

The key findings of this industrial investigation of TCR in the glass-melting furnace were the following:

- TCR using leads to fuel consumption decreasing;
- Waste-heat recuperation degree is almost two times higher for the furnace with TCR in comparison with the furnace without TCR;
- for high-temperature waste flue gases the TCR systems can be combined with the recuperation systems by air heating.

In 1986 the Dutch VEG-Gas Institute and TNO presented the results of the industrial application of the thermochemical waste-heat recuperation system in the glass-melting furnace [71,72]. The schematic diagram of a glass furnace is presented in Fig. 5.

According to this schematic diagram, flue gas after the glass furnace with a temperature level of about 1400° is consistently passed through recuperator, reformer, preheater of gas/steam mixture, and boiler. The intercooled flue gas after recuperator is used to heat a reaction space of a reformer where steam/gas mixture is reformed into synthesis gas. In this scheme, the natural gas reformer is a convenient cage-type heat exchanger in which the reaction space is filled with the catalysts. In the reformer, the enthalpy of flue gas is converted into sensible enthalpy of the synthesis gas and into chemical energy (in the form of increased

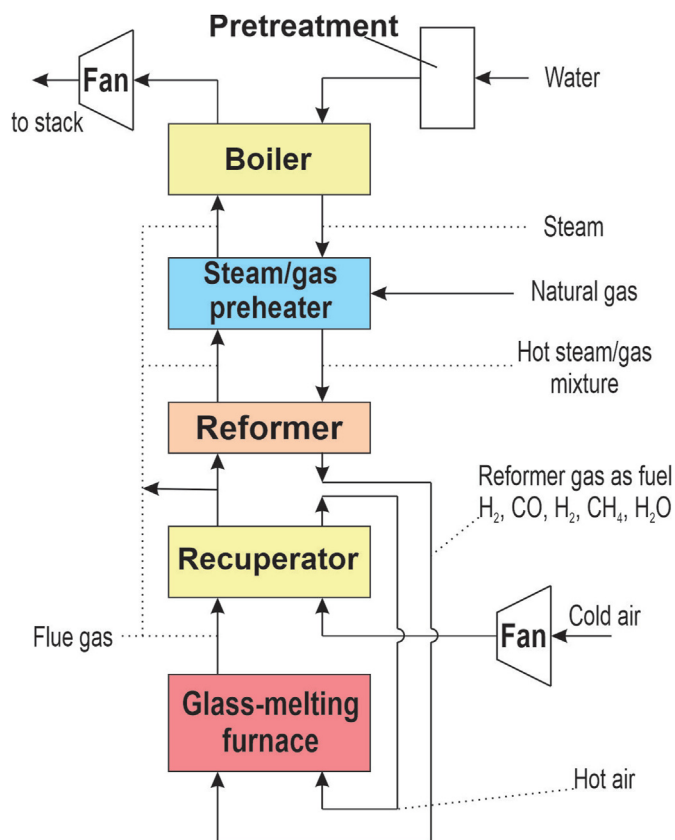


Fig. 5. The schematic diagram of a glass furnace with TCR developed by Dutch VEG-Gas Institute and TNO [71,72].

LHV in comparison with the original fuel). The hot air (700...800 °C) after the recuperator and hot synthesis gas (550...700 °C) are fed into the glass furnace.

The glass furnace with the thermochemical recuperation system (according to Fig. 5), as the authors reported in Refs. [71,72], has the following advantages:

- 65–70% of waste flue gas heat is recovered;
- part of the oxygen is delivered by steam instead of air, which leads to lower nitrogen concentration in the flue gases;
- application of the TCR system in the glass furnace with a capacity of 250 tons of glass per day leads to energy saving of 10–15%.

In the two mentioned above the industrial applications of TCR, the reformers were presented as the recuperative heat exchangers.

Table 3

The results of industrial testing of a glass-melting furnace without TCR and with TCR.

Parameter	Without TCR	With TCR		
		1	2	3
Gas consumption, m ³ /h	140	115	116	102
Air consumption, m ³ /h	1500	1185	1280	1150
Steam consumption, kg/h	—	162	190	157
Temperature of hot air, °C	725	365	425	610
Temp-re of steam-air mixture, °C	—	380	340	310
Temperature of syn.gas, °C	—	917	883	858
Methane conversion, %	—	90	79	65
Waste-heat recuperation degree	0.332	0.550	0.558	0.621

In Mexico, the regenerative thermochemical heat recovery system for oxy-fuel glass furnace with a capacity of 50 tons per day was realized at Pavisa (Mexico) [73,74]. The thermochemical regenerative heat recovery process patented by Kobayashi [75] was industrially utilized. The authors suggested using the methane reforming process for TCR. According to the authors of the project, the use of TCR will make it possible to reduce fuel consumption by about 30% compared to using only thermal recuperation (air heating). OPTIMELT™ thermochemical recuperation technology is used to recover waste flue gas heat by natural gas reforming. According to the authors' report [73], the OPTIMELT™ thermochemical recuperation system is simple and operated at atmospheric pressure without the catalyst for the natural gas reforming process.

Pure steam is a widely used oxidizing agent for thermochemical recuperation. However, flue gases after a gas furnace mainly consist of nitrogen, carbon dioxide, and steam. According to Table 1, both steam and carbon dioxide can be used as the oxidizing agent. In 1982 in USSR the research group led by Nosach constructed a heat-treating furnace with TCR [34,76]. In this TCR system, the natural gas was reformed with flue gases that contained steam and carbon dioxide. The schematic diagram of this thermochemical waste-heat recuperation system is presented in Fig. 6. A distinctive feature of such TCR system is no need to produce an oxidizing agent. Steam and carbon dioxide is used for the natural gas reforming process. The results of the industrial test of this heat-treating furnace are presented in Table 4. According to Table 4, in comparison with a furnace without TCR, a furnace with TCR and thermal recuperation by air heating provided an increase in fuel utilization factor from 54.1% to 91.5%. These results allow us to conclude that TCR provides almost complete waste-heat recuperation.

The industrial operation of the TCR systems shows its high efficiency. However, during operation, various problems arise, such as catalyst poisoning, burnout of the reformer wall, etc. Therefore, thorough testing of the TCR systems in laboratory conditions is an important task before industrial operation.

4.2. Laboratory tests

The specialists from GTI – Gas Technology Institute (Rue, Kozlov, Khinkis, Kurek) reported about thermochemical recuperation as a way for energy efficiency increasing of the glass furnaces [77]. In this report, Rue et al. presented the results of the application of the TCR system. They presented the energy and economic analysis of small-scale oxy-glass melter with a capacity of 5 tons per hour. These results were compared with experimental data. Pilot-scale TCR testing results were slightly lower than the simulation results. Moreover, the authors presented a photo of a pilot-scale TCR system developed by GTI in partnership with experts from Thermal Transfer, Inc. The main element of the TCR system is the tubular

Table 4

The results of industrial testing of a heat-treating furnace without TCR and with TCR by using flue gas for natural gas reforming [76].

Parameter	Without TCR	With TCR
Natural gas consumption, m ³ /h	22.7	13.0
Furnace capacity, kg/h	103	187
Temperature of hot metal, °C	1000	1000
Temperature of flue gas after furnace, °C	1060	1050
Temperature of exhaust gas, °C	1060	200
Temperature of hot air, °C	—	660
Temperature of syn.gas, °C	—	640
Fuel utilization factor, %	54.1	91.5
Heat consumption per 1 kg of metal, kJ/kg	7571	2388

recuperative methane reformer. The pilot-scale TCR system was integrated with a laboratory furnace. The experiments and simulations showed that overall fuel savings from TCR are from 25 to 30% relative to a recuperation system based on air preheating. The several performances of this pilot-scale TCR are summarized below:

- inlet exhaust gas temperature 1040 °C;
- exhaust gas flow rate up to 225 Nm³/h;
- reformed fuel flow rate up to 85 Nm³/h;
- reforming temperature 760 °C;
- furnace energy input 150 kWh.

The color photo of a laboratory furnace (Fig. 7) with TCR was presented at 74th Glass Problems Conference (October 15, 2013, Columbus, OH, USA) by Rue et al. [78]. In this laboratory furnace to simulate furnace loading, a water-cooled load is used. The TCR system is operating by using the steam methane reforming process.

Sikirica et al. [79] reported the experimental results obtained on a thermochemical recuperation test unit. The authors determined that the thermal efficiency of the furnace with TCR is increased by 15–35% and reduce natural gas consumption by 15–60% compared with the conventional thermal recuperation system when usually only combustion air is preheated. Moreover, the authors showed that the use of TCR will also significantly reduce by 30–80% air pollutant emissions. Sikirica et al. concluded that thermochemical recuperation due to steam methane reforming can be used for a wide range of industrial furnaces. The authors of this paper state: “We expect TCR to become a significant option for industry to consider in the future when addressing energy efficiency, and that GTI will play a major role in helping bring the technology to the market.”

A set of articles on the experimental investigation of thermochemical recuperation efficiency in lab-scale was published by specialists from the Graz University Of Technology. Gaber and co-

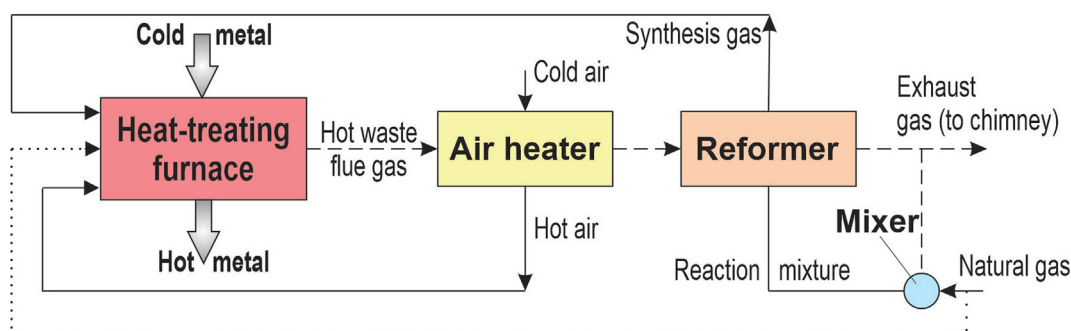


Fig. 6. The schematic diagram of a heat-treating furnace with TCR by natural gas reforming with flue gas [34,76].

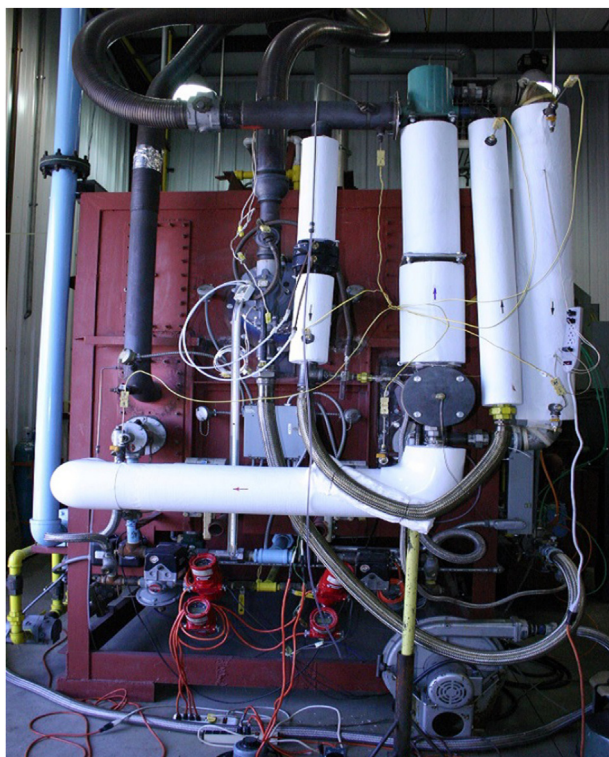


Fig. 7. Photo of pilot-scale TCR system with using steam methane reforming process [78].

workers investigated thermochemical recuperation by natural reforming [80–83]. Gaber et al. performed an experimental investigation of thermochemical regeneration using oxy-fuel exhaust gases [81]. The authors investigated the performance of steam-dry methane reforming (bi-reforming) and steam-dry reforming with oxygen addition of methane (tri-reforming). The principal difference of the investigated processes from the papers discussed above is using oxygen for methane combustion. Therefore, the exhaust gas consists of mainly steam and carbon dioxide, which are two main oxidizers for thermochemical recuperation. The schematic diagram of an experimental setup is presented in Fig. 8. Gaber and co-authors tested a wide range of practically interesting cases for the methane reforming process. They tested methane reforming with exhaust gas, exhaust gas with steam

addition, exhaust gas with oxygen, and exhaust gas with both steam and oxygen addition. The main attention in this work [81] was paid to the study of reforming characteristics and carbon formation conditions.

The main goals of this paper [81] were an investigation of the reforming characteristics and long-term stability of a Ni-based catalyst. The authors determined the stability of the methane reforming process for TCR application. The long-term stability of the thermochemical recuperation process is strongly limited by the carbon deposition on the catalyst particles. The main conclusions made by the authors are as follows: at the low oxidizer/methane ratio (less or equal to 1.0) the massive carbon deposition is observed; the addition of oxygen to the reaction mixture leads to a significantly increasing in the methane conversion and decreasing in the carbon deposition; a TCR unit had a stable operation with the constant reforming characteristics for at least 1350 min.

The scientific group from the Graz University Of Technology mentioned above based on the experimental results determined the waste heat recovery in a stationary operated thermochemical reformer [83]. Wachter et al. investigated the TCR system where oxidizers (steam and carbon dioxide) for the methane reforming process were taken from exhaust gases. They determined that the efficiency was increased 13.4% and 17.9%, respectively, during thermochemical exhaust heat recuperation. Also, they reported that for achieving the maximum waste heat recovery, the recirculation rate of exhaust gases should be set as high as possible.

The thermochemical recuperation concept can be used for the recovery of heat with the molten slags after the blast furnaces. In 1997 Kasai et al. [84] proposed the use of hot molten slag for the steam methane reforming process. The authors focused on the experimental determination of the kinetic characteristics of the steam methane reforming process as well as carbon deposition on the catalyst surface. Also, they investigated the slag composition on reforming characteristics of the methane reforming processed. A schematic diagram of the concept for thermochemical heat recovery system from molten slag is presented in Fig. 9. In this diagram, steam for steam methane reforming is fed after the slag granulation process. Moreover, hot synthesis gas is used for steam generation in a steam generator. The main findings of this paper are the following:

- the rate on the slag samples without sulfur addition is affected by the concentration of both methane and steam of the inlet reaction mixture. However, the concentration of steam becomes insignificant at a methane-to-steam ratio of less than 0.25;

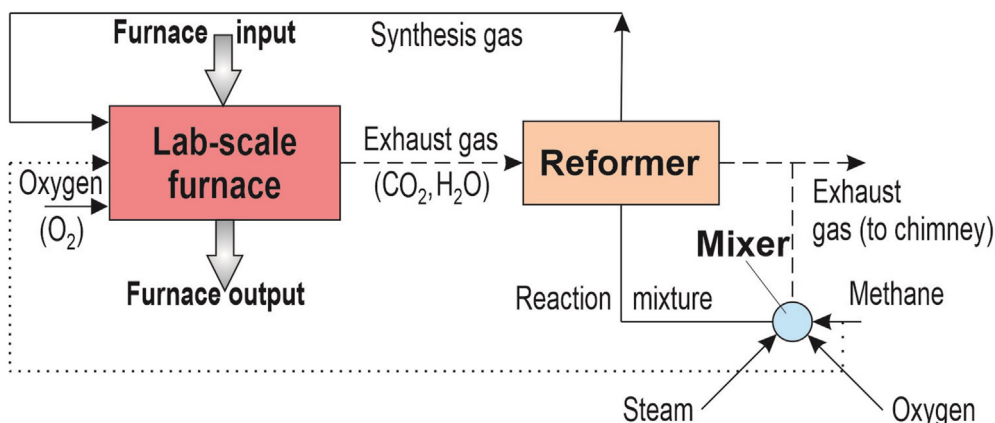


Fig. 8. An experimental setup for investigation of bi- and tri-reforming of methane [81].

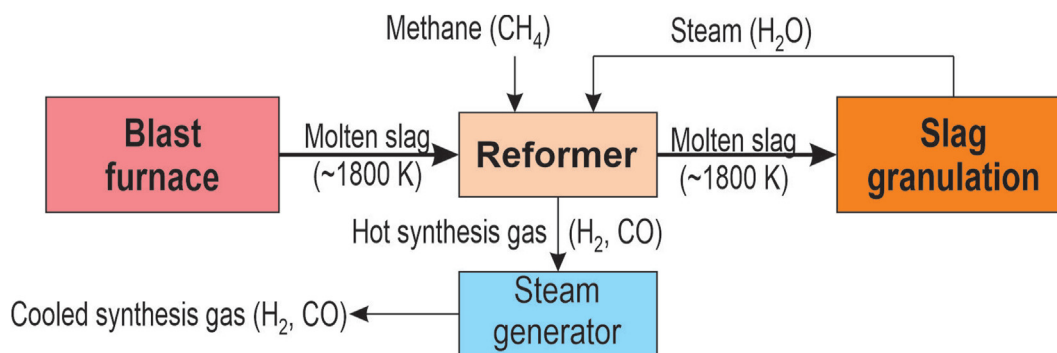


Fig. 9. A schematic diagram of the concept for thermochemical heat recovery system from molten slag [84].

- methane decomposition is a rate-limiting stage when slag containing a certain amount of sulfur or at a lower methane-to-steam ratio in the inlet reaction mixture.

In the continuation of the use of the slag heat, the work of Purwanto and Akiyama can be distinguished [85]. In this paper, the authors experimentally investigated the hydrogen production from the dry methane reforming process (CO_2 and CH_4) using the heat of the molten slag. The authors highlighted that the hot slags are playing a role as not only heat sources but also a good catalyst for methane reforming with carbon dioxide. Purwanto and Akiyama concluded that a new energy-saving process of hydrogen production based on a thermochemical transformation concept, in which the waste heat from molten slag can replace the energy required for an endothermic reaction, reducing carbon dioxide emission and increasing the energy efficiency of the hydrogen production process.

One of the interesting ways of TCR is a thermochemical transformation of nuclear heat into chemical energy via the steam methane reforming process. Firstly, this way of thermochemical transformation was experimentally investigated in Germany in 1975 [86]. Fedders et al. tested in laboratory ADAM/EVA scheme in which the heat with gas that cooling nuclear reactor was converted into chemical energy by the steam reforming reaction (EVA). In their experiments, This steam/methane mixture was fed to the reformer with a temperature between 400 and 550 °C and with a pressure of 25 and 35 bar. The authors mainly focused on the investigation of the steam methane reforming process from point of view of the potential for thermochemical transformation of the heat of the cooling gases. In 1975 the authors reported that the incorporation of nuclear heat into this process still requires comprehensive experimental and theoretical studies before an economic and technical optimization of a combined nuclear/chemical plant can be reached. Since then, many aspects have been studied in detail and thermochemical transformation due to methane reforming is widely discussed in industry and science.

4.3. Simulation analysis

Most of papers on the thermochemical waste-heat recuperation system by natural gas or methane reforming resented the results of the simulation analysis. The articles on TCR covers applications in various fuel-consuming equipment, among which two main types can be distinguished – industrial furnaces and gas turbines.

4.3.1. Furnaces with TCR by methane reforming

Thermodynamic analysis is a widely used method for simulation analysis. A distinctive feature of the thermodynamic analysis of the thermochemical waste-heat recuperation system is a simulation of

the chemical processes in the reformer. The equilibrium synthesis gas composition after the reformer depends on temperature, pressure, and oxidizer-to-fuel ratio. Maruoka et al. [87] presented the feasibility study for recovering waste-heat in the steel-making industry using a chemical recuperator. The authors proposed a thermochemical recuperation system for using hot wastes generated by the steel-making industry. In the suggested system, the exhaust heat was stored using a Phase Change Material, and after that used for an endothermic steam methane reforming process as a heat source. The authors calculated a heat and material balance based on the developed model. This model made it possible to predict the energy efficiency. The authors calculated that the exergy loss in the furnace with the thermochemical waste-heat recuperation system was only 15% from the total exergy losses in the furnace with the conventional recuperation systems. Moreover, Maruoka and co-authors determined the environmental and economic impact of TCR. They reported that the decrease in CO_2 emission due to TCR system is about 2.05 million tons per year for the steel-making industry of Japan; the cost benefits incurred by the use of TCR systems be steam methane reforming is about US\$ 409 million and US\$ 1945 million per year for discharged gas and slag, respectively.

Thermodynamic analysis of the thermochemical recuperation includes the processes which are taking place in various elements: the chemical reactions in the reformer, steam generation in the steam generator, steam-methane mixture preheating. For thermodynamic analysis of such large systems, special software can be used, for example, Aspen HYSYS or Aspen Plus. Popov et al. [21] presented the results of the thermodynamic analysis of different schematic diagrams of thermochemical recuperation in an industrial gas furnace. The thermodynamic analysis was performed with the help of Aspen Plus and MathCad. In this paper [21], the authors analyzed the TCR system based on natural gas reforming with steam flue gases and steam-flue gases mixture. The calculations were performed for a glass furnace with a capacity of 250 tons per day at an exhaust gas temperature of 1500 °C. Popov et al. determined that for TCR by steam reforming of natural, the fuel consumption is decreasing up to 34% compared to conventional recuperation based on air preheating to 800 °C. Moreover, for the glass furnace with TCR by natural gas reforming with steam, the fraction of recuperated waste-heat (recuperation degree) is 0.74. For the glass furnace with the recuperation system based on air preheating up to 800 °C the fraction of recuperated waste-heat is 0.38. Therefore, the authors concluded about the efficiency of TCR system based on steam reforming of natural gas.

For the scheme of TCR by natural gas reforming with flue gases, Popov et al. [21] analyzed the effect of oxygen mole fraction in the air for combustion. An increase in oxygen concentration leads to a decrease in the concentration of nitrogen in combustion products

(flue gases) that are used for natural gas reforming. Moreover, the authors analyzed the effect of the volume fraction of recirculated flue gases on the energy consumption per kg of glass. The specific energy consumption is minimum in the range of volume fraction of recirculated flue gases from 0.21 to 0.25. For atmospheric air with an oxygen volume concentration of 0.21, the maximum efficiency is for the volume fraction of recirculated flue gases of 0.23.

In the same article, Popov et al. [21] investigated the efficiency of TCR system based on natural gas reforming with the mixture of flue gases and steam. The authors showed that an increase in steam addition to a flue gas leads to decreasing in optimal volume fraction of recirculated flue gases from 0.19 to 0.22 (depending on the volume of added steam).

The results of comparison of the specific energy consumption per 1 kg of glass for various schematic diagrams of the TCR systems (when atmospheric air is used) are presented below:

- TCR based on natural gas reforming with steam 4.73 MJ/kg (steam-to-natural gas ratio of 2.0) and 4.97 MJ/kg (steam-to-natural gas ratio of 3.0);
- TCR based on natural gas reforming with flue gases 5.21 MJ/kg (volume fraction of recirculated flue gases of 0.23);
- TCR based on natural gas reforming with the mixture of flue gases and steam 4.48 MJ/kg (0.09 m³ of steam per 1 m³ of natural gas) and 4.44 MJ/kg (0.21 m³ of steam per 1 m³ of natural gas)

The thermodynamic analysis performed by Popov et al. compared various schematic diagrams of the TCR system and for given conditions. The minimum specific energy consumption is observed for the TCR system based on natural gas reforming with the mixture of flue gases and steam.

The author of this review also performed an analysis of various TCR systems in industrial furnaces. The detailed first law energy analysis of thermochemical waste-heat recuperation by steam methane reforming was carried out based on the thermodynamic analysis [88]. Aspen HYSYS and IVTANTHEMRO [89] were used for the calculation of equilibrium synthesis gas composition and enthalpy of the reforming process. The heat balances of TCR system were calculated for various operational parameters: temperature and the steam-to-methane ratio at the reformer inlet. This first law energy analysis made in the possible determination of operation conditions for effective using of thermochemical recuperation of waste flue gases: temperature range 900–1100 K for the steam-to-methane inlet ratio of 2 and temperature range above 1200 K for the steam-to-methane inlet ratio of 2; optimal pressure is 5–10 bar. Moreover, this article highlighted one of the main disadvantages of the TCR systems based on the steam methane reforming process – the need to generate steam. Heat consumption for steam production is from 8 to 16% of the total heat in the furnace for a steam-to-methane ratio of 1 and 2, respectively.

One way to overcome the disadvantage associated with steam production is using flue gases for methane reforming. Flue gases contain steam and carbon dioxide, which are used to reform methane. The author's previous work is dealing with the thermodynamic analysis of thermochemical waste-heat recuperation by steam methane reforming with flue gas addition [33]. Aspen HYSYS was used for the simulation analysis. The calculation Aspen HYSYS model of thermochemical waste-heat recuperation by steam methane reforming with flue gas addition is presented in Fig. 10.

In this paper, the main advantages of such way of thermochemical recuperation are highlighted:

- reducing the heat input in a steam generator for a steam generation;

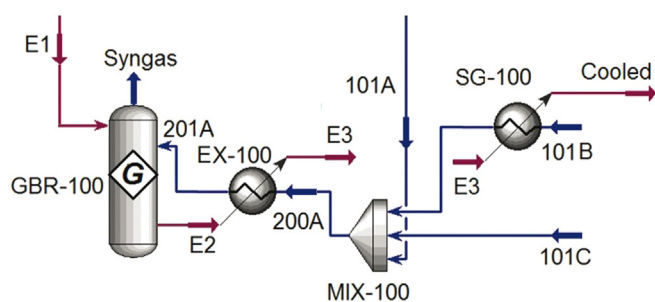


Fig. 10. Aspen HYSYS process flow diagram for thermochemical recuperation (TCR) by steam methane reforming with addition of combustion products to the reaction mixture [33]: GBR-100 – Gibbs' reactor (methane reformer); EX-100 – preheater of the steam-methane mixture; MIX-100 – mixer of steam and methane; SG-100 – steam generator; E1 – hot exhaust gas; E2 – first intercooled exhaust gas; E3 – second intercooled exhaust gas; Cooled – exhaust gas to stack; 101C – methane (CH₄); 101B – water (H₂O); 101A – combustion products (H₂O, CO₂, N₂); 200A – unheated steam-methane mixture; 201A – hot steam-methane mixture.

- reduction of the steam generator size due to the need to produce a smaller amount of steam in comparison with TCR by pure steam methane reforming;
- reducing the amount of heat transferred through the wall of the reformer, and, as a consequence, reducing the specific amount of metal for the reformer.

Thermodynamic equilibrium analysis makes it possible to establish the effect of operational parameters such as temperature, pressure, and reaction mixture composition at the reformer inlet on the efficiency of TCR system. However, to determine the effect of reformer design, type of catalysts, shape of catalyst, we need to use CFD-modelling (numerical simulation). The articles on a numerical simulation of the methane reforming processes have a wide variety [90–95]. In respect to numerical simulation of the thermochemical waste-heat recuperation process, Zhang et al. performed the numerical analysis of tubular reformer where a heat source for methane reforming process is supplied from the exhaust gases [96]. The authors analyzed TCR based on methane reforming with a mixture of steam and combustion products. The results of this paper are the following: TCR based on methane reforming can be considered as on-board hydrogen production technology; the hydrogen production increases first and then decreases with the increase of exhaust gas reforming ratio, peaking when the methane-to-oxygen ratio is about 2 regardless of the excess air coefficients; with all other parameters fixed, there is an optimum amount of steam addition to achieving the maximum hydrogen production, which occurs when the steam-to-methane ratio is about 1.25.

4.3.2. Gas turbines

The modern gas turbine units have a temperature of combustion products before a turbine above 1700–1800 K. Therefore, the exhaust gas temperature can reach about 800–1000 K. Such a temperature potential is enough for carrying out the methane reforming processes for thermochemical recuperation of exhaust heat. The schematic diagrams with.

As discussed above, Olmsted and Grimes were the first who proposed using thermochemical recuperation based on steam methane reforming for the gas turbines [31]. The authors suggested partially replacing of steam superheater section with a steam methane reformer. The exhaust heat after a gas turbine was used for the endothermic steam methane reforming process. Then the products of reforming are fed into the turbine combustor. In that paper, Olmsted and Grimes showed that CH₄/H₂O absorbs exhaust

heat thermally (as the temperature of mixture increases), and chemically (as the endothermic process takes place), resulting in a higher potential recuperation of exhaust heat than can be observed by conventional recuperation using air heating. Further, the idea of using TCR in gas turbines was found a wide discussion in the scientific literature. A detailed analysis of the gas turbines with TCR will be presented below.

The schematic diagrams of the GTU with thermochemical recuperation systems are very similar to the schematic diagram of the furnaces with TCR. However, the papers on the application of TCR in the GTU have a wide variety. The principal schematic diagram of GTU with TCR by steam methane reforming is presented in Fig. 11. This schematic diagram can be virtually divided into two subsystems: the power generation subsystem (compressor, combustion chamber, gas turbine, generator) and the thermochemical exhaust heat recuperation subsystem (reformer (SMR) and steam generator (SG)).

Various schemes of TCR implementation in the GTU were proposed in last years [97–101]. Kesser et al. performed the thermodynamic analysis of a gas turbine power plant with thermochemical recuperation via steam methane reforming [102]. The efficiency of a gas turbine with TCR was compared with a steam-injected gas turbine, a combined cycle, and a simple gas turbine without any recuperation system. The authors determined that the gas turbines with TCR with a compressor pressure ratio of 30 have a thermal efficiency of about 47.0% while for a pressure ratio of 15, the thermal efficiency is about 47.3%. Both these variants are attractively high when for a steam-injected gas turbine, a combined cycle, and a simple gas turbine without any recuperation system.

Verkhivker and Kravchenko carried out the thermodynamic analysis of a combined cycle plant with thermochemical recuperation based on steam methane reforming [11]. The authors proposed using steam for reforming from a high-pressure cylinder of a steam turbine. Verkhivker and Kravchenko calculated the energy efficiency of a combined cycle plant with thermochemical recuperation has reached up to 90%. Additionally, the authors showed the positive impact of thermochemical recuperation on the environment. The use of synthesis gas after TCR leads to decreasing the amount of waste carbon dioxide ejected into the environment by nearly 20%.

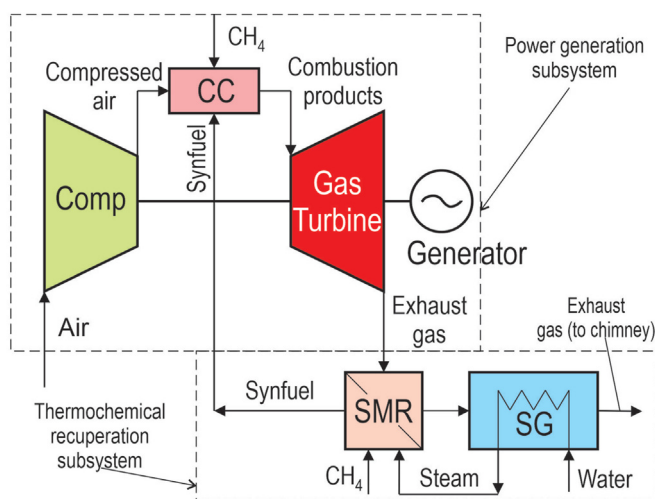


Fig. 11. A schematic diagram of GTU with TCR via steam methane reforming: Comp – compressor; CC – combustion chamber; SMR – steam methane reformer; SG – steam generator.

Rostrup-Nielsen proposed to use the steam reforming reaction with the high endothermicity for “chemical” recuperation of high-temperature heat [103]. Rostrup-Nielsen showed that thermochemical recuperation is an alternative for a convenient way of using hot exhaust gas by steam generation for the steam turbines.

Sadeghi et al. presented the results of thermoeconomic comparison and multi-objective optimization of combined gas turbine and organic Rankine cycle power generation system with thermochemical recuperation based on steam methane reforming and thermophysical recuperation [104]. The main goal of this paper is proposing and comparing (thermodynamically and economically) three power generation systems consisting of simple gas turbine/organic Rankine cycle, gas turbine/organic Rankine cycle with thermophysical recuperators, and gas turbine/organic Rankine cycle with thermophysical and thermochemical recuperators. The results showed that the thermodynamic, economic, and environmental efficiency of all systems is improved when the temperature of combustion products increases. The authors reported that the power generation system with thermophysical recuperators shows the best performance from the thermodynamic perspective with the 22.86% and 27.52% improvement in the exergy efficiency compared to the simple gas turbine/organic Rankine cycle and gas turbine/organic Rankine cycle with TCR, respectively. However, under the operating thermodynamic conditions obtained by the multi-objective optimization approach, the product cost associated with the system using TCR is less than that of the other configurations.

Cao and Zheng published a set of articles on TCR by carbon dioxide methane reforming for GTU [105,106]. In these papers, the authors proposed a GTU system with a thermochemical recuperative cycle by dioxide natural gas reforming. The thermodynamic analysis of a gas turbine cycle with thermochemical recuperation by carbon dioxide reforming of methane [106] showed that net power generation efficiency of the cycle rises to 49.6% and the exergy efficiency 47.9%. Aspen Plus was used for this thermodynamic analysis. Moreover, the author stated that using TCR leads to decreasing in pollutants such as NO_x and CO_2 . Another work of these authors presented the results of exergy analysis [105] of GTU with TCR by CO_2 -methane reforming. Cao and Zheng compared the performances of the gas turbines when methane and synthesis gas are used as fuel. Comparing the combustion of CH_4 (methane) and CO-H_2 mixture (synthesis gas), the combustion heat of the synthesis gas increases 37.8%, but more importantly, the exergy released increases 30.6% and the exergy losses for the synthesis gas decrease 25.5% at the same temperature of combustion.

Lui et al. presented the results of the investigation of methane reforming and combustion characteristics in thermochemically recuperated gas turbine [107]. The authors investigated the thermochemical recuperation concept from Fig. 11. The focus of this paper was forwarded to CFD-modeling of steam methane reforming and numerical investigation of synthesis gas combustion. The experimental investigation was performed to verify the results of the CFD modeling of the steam methane reforming process. Lui et al. showed the energy efficiency of using TCR in GTU. Moreover, they compared the combustion characteristics of methane and synthesis gas in a diffusion flame. For the synthesis gas, the flame is shortened and becomes narrow due to the inflammability of hydrogen. Also, the maximum temperature of flame drops because there is steam in the synthesis gas (for a steam-to-methane ratio of 2). The authors stated that TCR can be considered as ultralow NO_x emission technology.

Florez et al. [108] analyzed four proposed configurations based on the thermochemically recuperated gas turbine concept. The authors compared these configurations in terms of the minimum specific exergy destruction rate, the plantwide and cogeneration

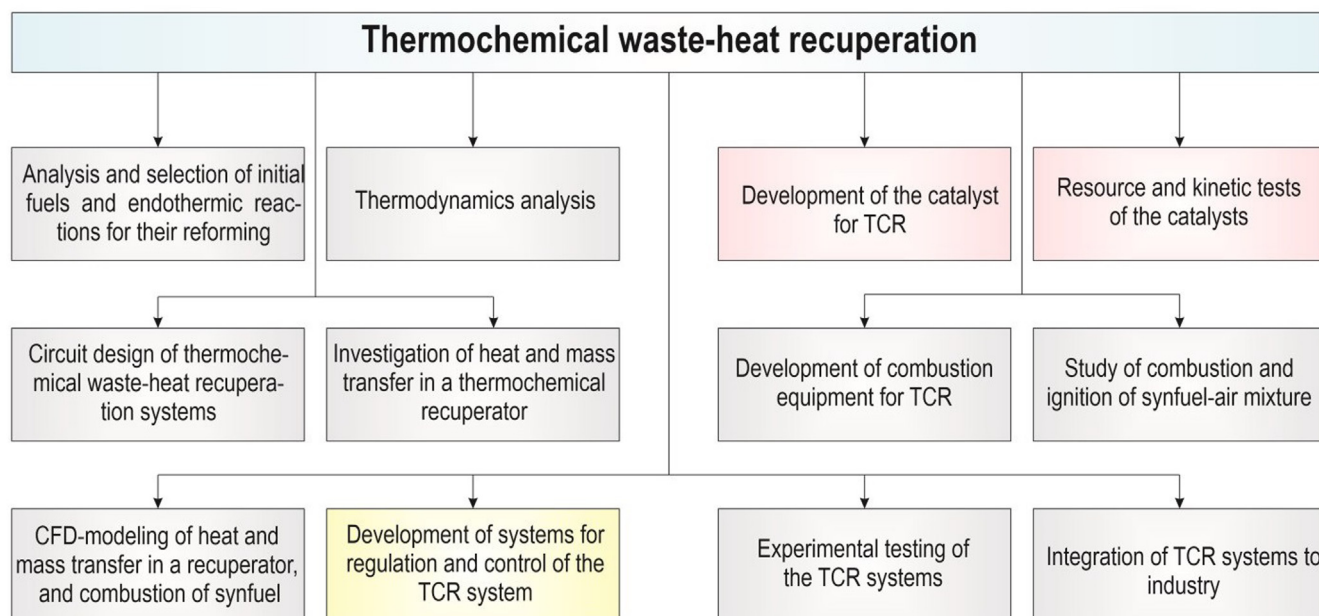


Fig. 12. The tasks for investigation of thermochemical recuperation.

exergy efficiency, and atmospheric carbon dioxide emissions. According to their analysis, the exergy fuel consumption in the proposed gas turbines with TRC has been estimated as 0.5–2.7% lower than in the conventional design. However, the most striking characteristic of the proposed GTUs with TCR has been their capability of cutting down up to 25% the specific atmospheric carbon dioxide emissions, compared to the conventional case.

4.3.3. Elements of thermal protection

TCR based on reforming of hydrocarbons can also find application in systems for thermal protection of thermally stressed elements. In this case, the heat of the thermally stressed element is converted into the chemical energy of the synthesis gas in a reformer. In this case, in addition to recuperating the heat of the thermally stressed element, intensive thermal protection of the thermally stressed element occurs due to heating of the heat carriers (reaction mixture of hydrocarbon fuel and an oxidizer, for example, steam) and due to the endothermic reforming process.

The first works in this way of TCR were published in the 1960s. Nixon et al. proposed using endothermic reforming of fuel before supplying it to the combustion chamber for increasing the cooling power of hydrocarbon fuels as a coolant [109]. Lander and Nixon [110] proposed using various reactions, such as thermal cracking, depolymerization, dehydrogenation, and dehydrocyclization for hypersonic vehicles.

Kuranov et al. proposed to use steam methane reforming and methane cracking for thermochemical recuperation of heat of the thermally stressed elements [111]. The authors considered the issues of increasing the cooling capacity of methane by means of its thermochemical transformation in the thermal protection elements of a hypersonic aircraft and obtaining hydrogen on board for the organization of stable combustion in a hypersonic ramjet engine. Also, they performed a comparison of the chemical reactions of cracking and steam methane reforming in terms of the value of the cooling resource and the amount of hydrogen in the reaction products is carried out.

5. Research tasks of TCR

Above, the use of TCR based on methane reforming in industrial furnaces and gas turbines was considered. Moreover, the works that deal with experiments, numerical modeling, and thermodynamic analysis of TCR were analyzed. In most of these works, a positive energy effect from the use of TCR is noted. In addition, some works also investigate the emission characteristics of the use of synthesis gas after the TCR system as fuel and note a positive effect on the emissions of NO_x and CO_2 .

Analyzing the work on TCR (not only based on methane reforming), the author has developed a flowchart for solving problems of the practical application of TCR. This flowchart can be virtually divided into six blocks:

- thermodynamic analysis;
- TCR configuration;
- heat and mass transfer in the reformer;
- catalysts design;
- synthesis gas combustion;
- experimental testing and industry application.

A flowchart for solving problems of the practical application of thermochemical recuperation of waste-heat is presented in Fig. 12.

6. Conclusion

In this review, various configurations of thermochemical waste-heat recuperation systems developed for different fuel-consuming equipment have been overlooked. Therefore, the previous studies carried out on different aspects of these systems have been reviewed. A historical retrospective of the application of thermochemical recuperation from the 1960s to the present day has been provided. A classification of thermochemical recuperation systems is proposed. The type of fuel in TCR systems is an important characteristic for classification. A brief overview of TCR via the

reforming of liquid and solid fuels is presented. The main purpose of this article is to provide an overview of TCR based on the reforming of natural gas and methane. Insofar as the degree of methane conversion tends to unity at temperatures above 1000 K, the main area of application of TCR due to methane reforming is the recovery of high-temperature heat. In this regard, the main attention is focused on the use of TCR in industrial furnaces and gas turbines. The results of industrial use of TCR, experimental research, and simulation modeling are presented. For the first time, the results of industrial use of TCR in glass-melting furnaces, which were previously published only in Russian, are presented in English.

In this paper, the available articles, patents, books on thermochemical waste-heat recuperation based on reforming of natural gas (methane) were systematized and summarized. In this review, thermochemical recuperation of various types of waste-heat such as exhaust gas, molten slag, elements of thermal protection has been considered. In addition, TCR systems based on various reactions of methane reforming (with steam, carbon dioxide, a mixture of steam and carbon dioxide, steam and flue gases, a mixture of steam and oxygen) are considered. The industrial use of TCR in gas furnaces shows good energy efficiency. On average, the efficiency of a furnace with TCR increases by 20–30% in comparison with a traditional heat recovery system based on air heating. In addition, the use of TCR in combined cycles allows not only to increase efficiency but also to simplify the design of such cycles. In some works, it is noted that TCR leads to a decrease in the exergy efficiency of the GTU, while the overall economic and energy efficiency increases.

The analyzed TCR systems perform not only the traditional function of ensuring heat recovery but also allows the production of fuel with superior characteristics containing molecular hydrogen. As a result, it becomes possible to use not only primary hydrocarbon fuel but also hydrogen-rich gas – products of endothermic conversion. Taking into account the global trend of using hydrogen as an energy source, the TCR system with endothermic hydrocarbon fuel reforming could be utilized for producing hydrogen at a plant where it is consumed (on-board hydrogen production system). According to this statement, the impact of TCR on the emission of NO_x and CO_2 , when synthesis gas after TCR systems are using, have been analyzed. TCR based on steam methane reforming can be considered as ultralow NO_x emission technology for GTU application.

After analyzing the articles on thermochemical recuperation, a flowchart of tasks for the study of TCR was proposed. This flowchart can be virtually divided into six blocks: thermodynamic analysis; TCR configuration; heat and mass transfer in the reformer; catalysts design; synthesis gas combustion; experimental testing and industry application. The development of modern tools for CFD-modelling and thermodynamic analysis makes it possible to use the principles of thermochemical recuperation for various fuel-consuming equipment.

While many studies have been conducted in the field of thermochemical recuperation and on-board hydrogen fuel reforming, many issues still need to be focused for the future researches. In section (5) the research issues for TCR was systematized in a flowchart diagram. These issues can be virtually divided into six blocks: thermodynamic analysis; TCR configuration; heat and mass transfer in the reformer; catalysts design; synthesis gas combustion; experimental testing and industry application. Therefore, there are still numerous subjects that require further investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2022.123854>.

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